Quantum Teleportation

Explorer Demonstration

Pinakin Padalia

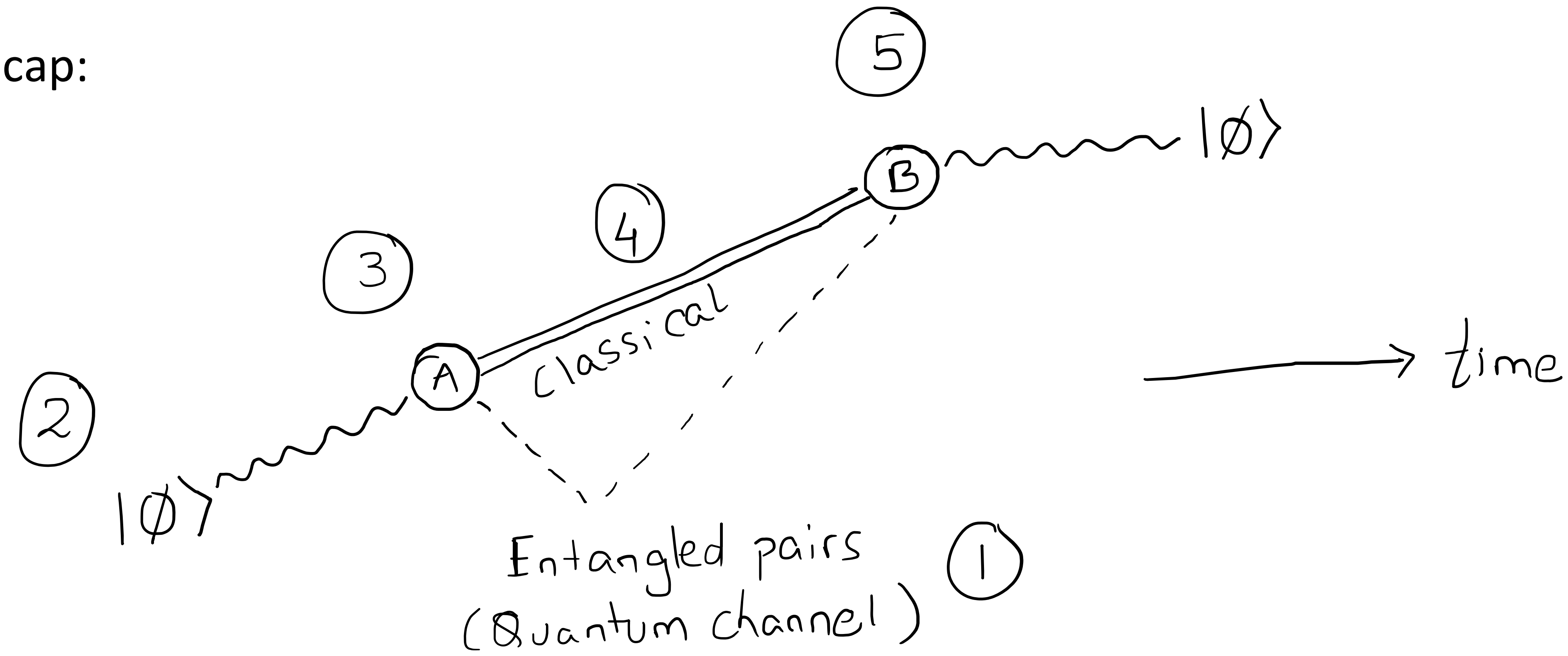
What are we covering in this chapter?

- Quantum Teleportation: What and why?
- Quantum Teleportation: How?
- **QPiAI Explorer demonstration**

Module 4.1

Teleportation Protocol

- Recap:

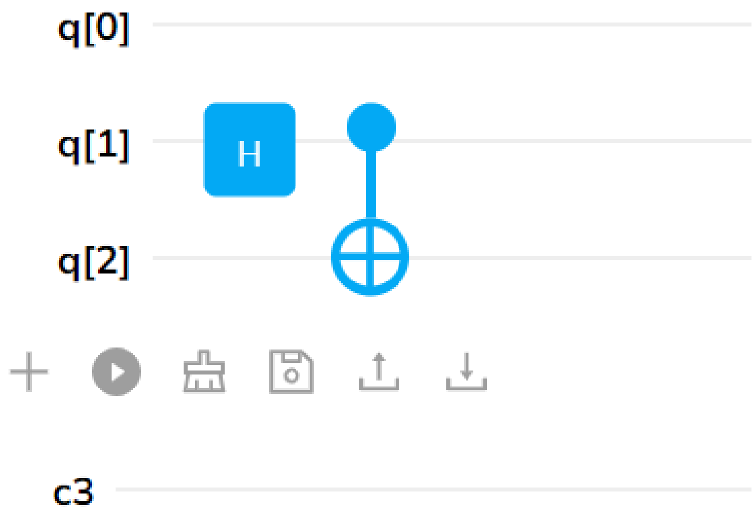


What are we covering in this chapter?

- Assumptions: 1st qubit is to be teleported, 2nd qubit is part of the bell pair with Alice and 3rd qubit is part of the bell pair with Bob.

- Step 1: Prepare the entangle pair

$$\frac{1}{\sqrt{2}}\left((|00\rangle + |11\rangle)\right)$$

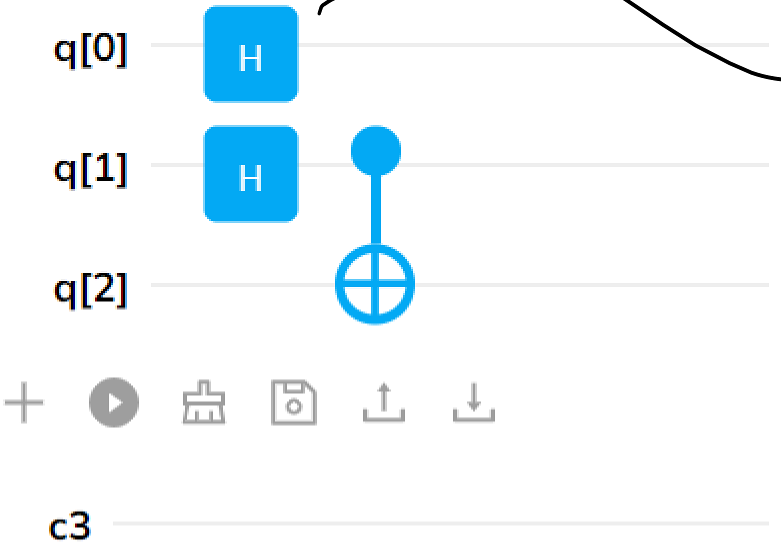


$$|00\rangle \xrightarrow{H} \frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

$$\xrightarrow{CNOT} \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

What are we covering in this chapter?

- Assumptions: 1st qubit is to be teleported, 2nd qubit is part of the bell pair with Alice and 3rd qubit is part of the bell pair with Bob.
- Step 2: Prepare the state to be teleported: $|+\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)$



$|0\rangle \xrightarrow{H} |+\rangle$

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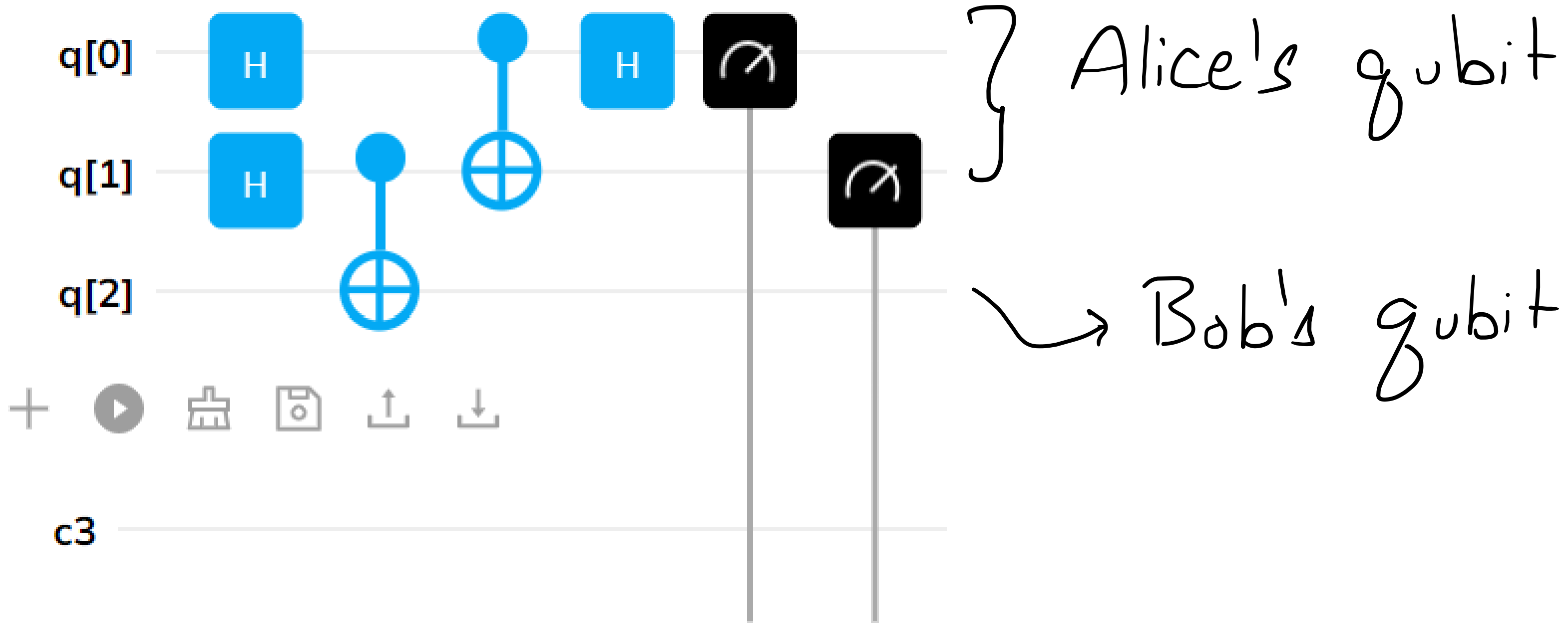
- Step 3: perform bell measurements:



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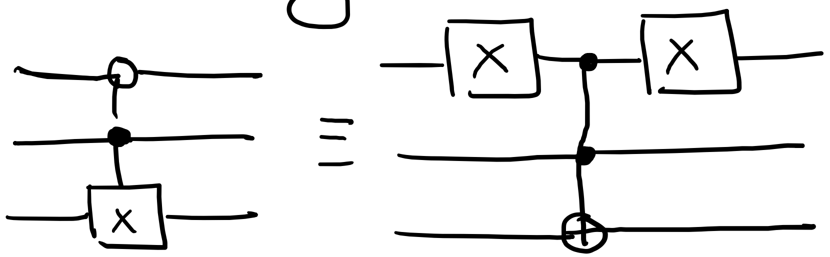
- Assumptions: 1^{st} qubit is to be teleported, 2^{nd} qubit is part of the bell pair with Alice and 3^{rd} qubit is part of the bell pair with Bob.

- Step 4/5: Perform operations on Bob's qubit based on outcomes

0 0 → I → Do nothing

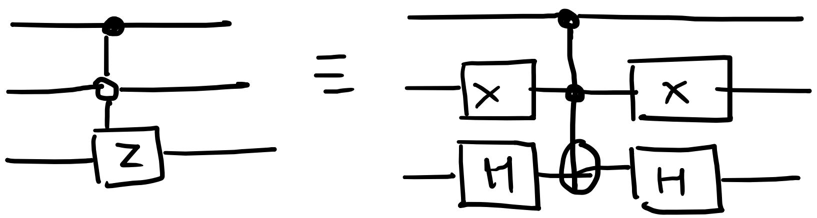
1 1 → X Z

0 1 → X →



combine

1 0 → Z →



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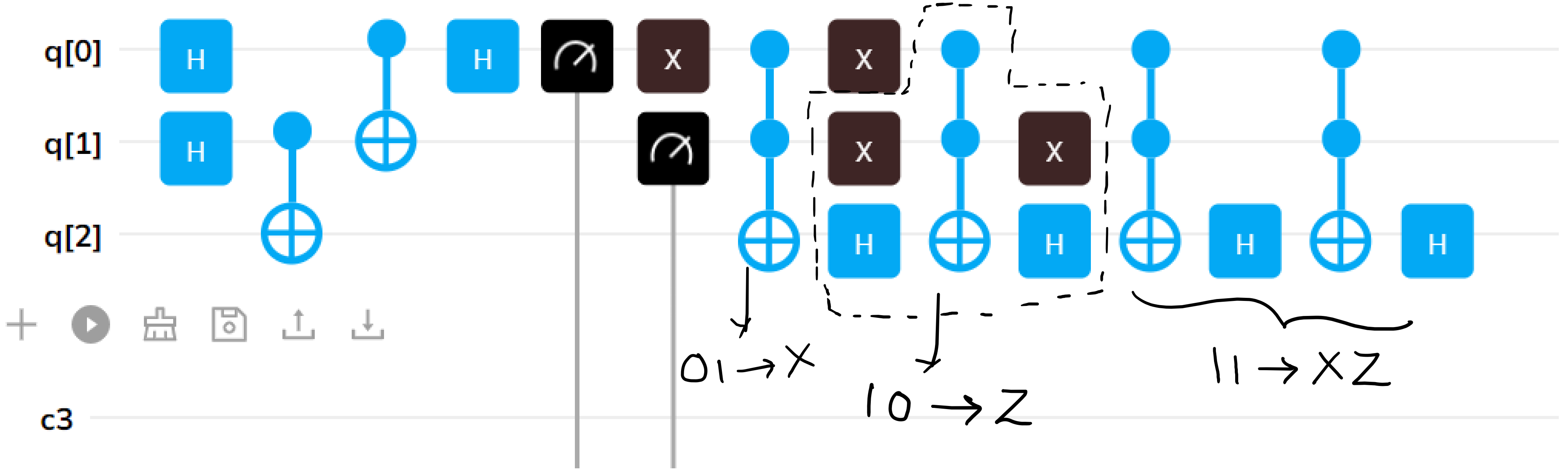

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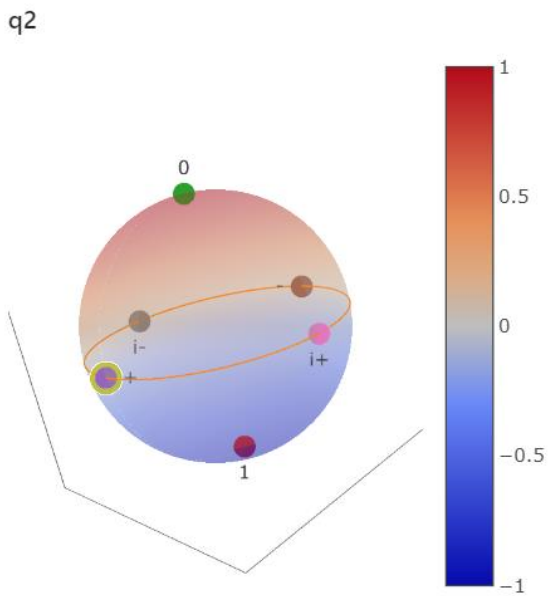

Bob's qubit will always be in $|+\rangle$ State.

Thank you for your attention!